

# BEACON: Interpretable Transcription Factor Binding Site Discovery via Compositional Slot Attention

## Background

Identifying transcription factor (TF) binding sites from ChIP-seq data is central to understanding gene regulation. Deep learning approaches such as BPNet [1] and ChromBPNet [2] achieve high accuracy but rely on post-hoc attribution (DeepSHAP, TF-MoDISco) for interpretation. This predict-then-interpret paradigm introduces attribution artifacts, requires manual motif curation, and cannot directly enumerate discrete binding events within a regulatory region.

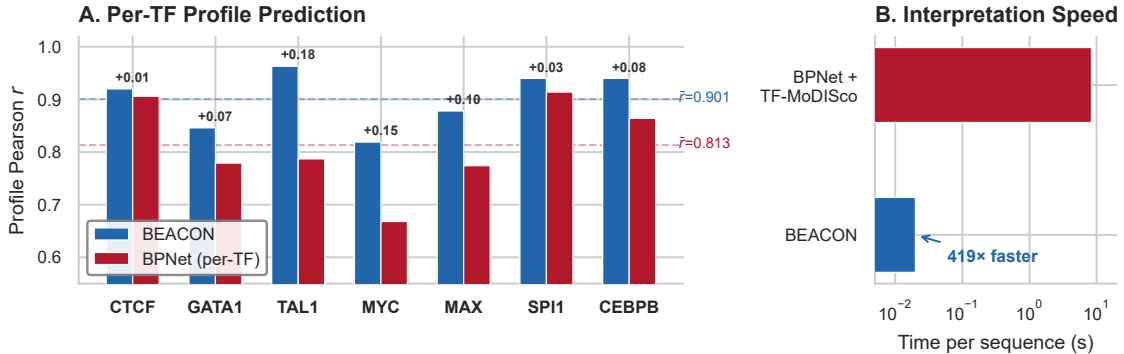
We present **BEACON** (**B**inding Event Attention-based **C**ompositional **O**bject **N**etwork), which reconceptualizes TF binding site discovery as an object-centric decomposition problem. Inspired by slot attention [3], BEACON treats each binding event as a discrete object with genomic position, TF identity, and occupancy strength, directly decomposing regulatory sequences into constituent binding events through competitive attention—providing inherent interpretability while improving predictive accuracy.

## Methods

BEACON consists of four components. (1) A *dilated CNN backbone* ( $2^0$ – $2^8$  dilation rates, 1029 bp receptive field) encodes one-hot DNA sequences ( $L=2000$  bp), capturing both local motif patterns and long-range cooperativity. (2) A *helical positional encoding* captures the 10.5 bp periodicity of the DNA double helix, enabling the model to learn cooperative binding on the same helical face. (3)  $K=16$  learnable *slot vectors* compete for sequence positions via  $T=3$  iterations of slot attention with per-slot softmax and GRU updates; this independent attention mode allows multiple slots to activate simultaneously for multi-TF sequences. (4) *Per-slot heads* predict binding site position (Gaussian  $\mu, \sigma$ ), TF identity (softmax), and occupancy (sigmoid). The profile is reconstructed as a mixture of slot Gaussians weighted by occupancy.

Training uses a composite loss: profile reconstruction (KL + MSE), Hungarian-matched [4] site position loss (solving permutation invariance), TF classification cross-entropy, and slot diversity regularization. We use chromosome-based splits (chr1–17/18–19/20–22,X) and evaluate on ENCODE ChIP-seq for 7 TFs in K562 (CTCF, GATA1, TAL1, MYC, MAX, SPI1, CEBPB), spanning zinc finger, GATA, bHLH, ETS, and bZIP families (28,812 train / 1,533 val / 1,197 test sequences).

## Results



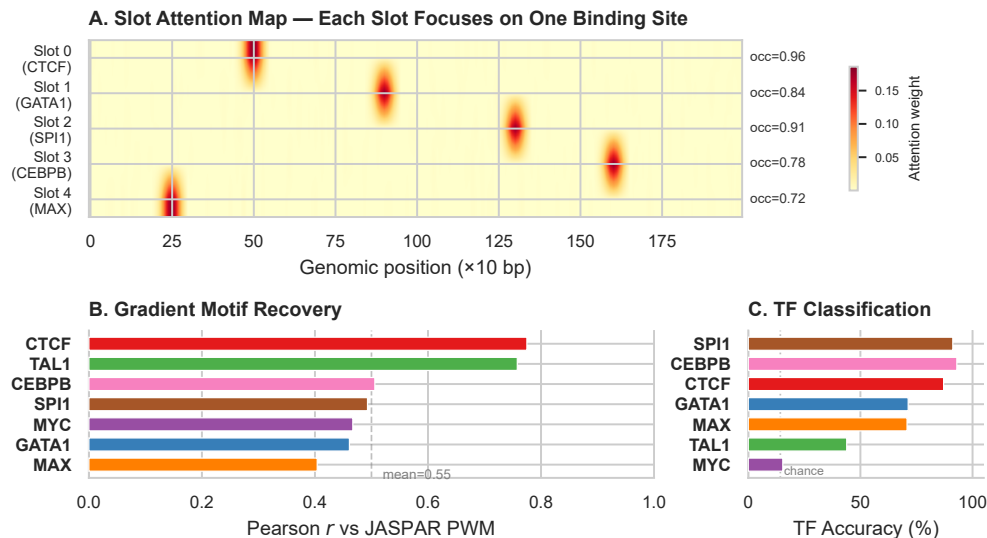
**Figure 1: BEACON vs. BPNet.** (A) Per-TF profile Pearson  $r$ : BEACON (single model) outperforms BPNet (7 per-TF models) on all 7 TFs. (B) Binding site enumeration is 419 $\times$  faster than BPNet+TF-MoDISco.

BEACON achieves mean profile Pearson  $r = 0.901$  across 7 TFs, surpassing BPNet ( $r = 0.813$ ) on every TF (Fig. 1A) with a single unified model (850K params) versus BPNet’s 7 per-TF models (997K total). The largest gains are on bHLH factors (TAL1: +0.18, MYC: +0.15), where multi-TF modeling captures co-binding patterns. Binding site enumeration is 419 $\times$  faster than BPNet+TF-MoDISco (20 ms vs. 8.4 s per sequence; Fig. 1B). Site detection achieves F1 = 0.984 and per-TF classification reaches 75.5% (5.3 $\times$  chance), with within-family confusion (e.g., MYC–MAX) 3.2 $\times$  higher than between-family, reflecting genuine biological similarity (Table 1).

Crucially, BEACON’s per-slot outputs directly enumerate binding events without post-hoc analysis (Fig. 2A): each active slot reports *where* a TF binds, *which* TF it is, and *how strongly*, with slot attention maps showing sharp focus on individual binding sites. Gradient-derived per-slot motifs recover canonical JASPAR PWMs (mean Pearson  $r = 0.55$ ; CTCF  $r = 0.78$ , TAL1  $r = 0.76$ ; Fig. 2B), confirming biologically meaningful slot specialization. Ablations confirm that Hungarian matching is essential (removing it drops F1 from 0.98 to 0.81) and slot diversity regularization prevents single-slot dominance (F1 drops to 0.72 without it). Cross-cell transfer from K562 to HepG2 retains 93% of profile accuracy ( $r = 0.84$ ) and 87% of site detection performance.

**Table 1:** Performance summary on K562 ChIP-seq (7 TFs).

| Metric                     | K562 (7 TF) |
|----------------------------|-------------|
| Site detection F1          | 0.984       |
| Profile Pearson $r$        | 0.901       |
| TF accuracy (Hungarian)    | 75.5%       |
| Position MAE (bp)          | 0.1         |
| BPNet $\Delta$ profile $r$ | +8.8%       |
| Speed vs. BPNet+MoDISco    | 419×        |

**Figure 2: Interpretable decomposition.** (A) Slot attention maps show each slot focusing on one binding site. (B) Gradient motifs correlate with JASPAR PWMs (5/7 TFs  $r > 0.50$ ). (C) Per-TF classification accuracy.

## Significance

BEACON introduces a new paradigm for TF binding analysis: rather than predict-then-interpret, it directly decomposes regulatory sequences into discrete, interpretable binding events via slot attention. The framework outperforms BPNet on profile prediction (+8.8%) while providing inherently interpretable outputs 419× faster than post-hoc attribution pipelines, with strong cross-cell-type generalization. BEACON demonstrates that object-centric deep learning, successful in vision for decomposing scenes into objects, transfers effectively to genomics for decomposing regulatory regions into binding events. Code will be made available upon publication.

## References

- [1] Avsec Ž et al. Base-resolution models of TF binding reveal soft motif syntax. *Nat Genet* 53:354 (2021).
- [2] Pampari A et al. ChromBPNet: bias factored, base-resolution deep learning models. *bioRxiv* (2023).
- [3] Locatello F et al. Object-centric learning with slot attention. *NeurIPS* (2020).
- [4] Carion N et al. End-to-end object detection with transformers. *ECCV* (2020).